Bournemouth University

School of Design, Engineering and Computing

Course:	**B.Sc. Computing**
	B.Sc. Software Engineering Management
Unit:	**Systems Architecture**
Assignment:	**One**
Due Date:	**24th January 2005**

Aims and Objectives

This assignment provides the student with the opportunity to demonstrate their ability to work in a small group, to practice their ability to view their own work, and that of their colleagues, in a critical light, in the process of demonstrating their understanding of circuit design techniques.

The assignment is intended to address the following learning outcomes:

2: Describe the relationship between the process, bus, memory and input-output devices

3: Understand the relationship between computer hardware and software

5: Explain how a computer system processes real-time events

The Task

You are to work in groups (of two or three) to design a time-of-day clock that can display the time from 00:00:00 to 23:59:59. You should assume an external clock signal will pulses twice a second. Note that using the 4-bit counter and BCD to 7-segment display decoder macros provided by Digital Works will simplify the design. To obtain additional marks your group may like to attempt either of the following, optional extensions:

- Have the second colon (":") (the one between the minutes and seconds display) flash on the half second.
- Alter the display to use the 12-hour clock with an AM/PM indicator.

Deliverables

The team is to produce a single "team" report containing the following:

- A Coursework Report sheet completed with the names of all the group members, and your lab group.
- A printout of your final circuit design.
- A description of the team's approach to the problem.
- Design documentation (truth tables, Boolean expressions, karnaugh-maps, sub-circuits, etc.) for those parts of the circuit that were created as a group
- A collection of individual reports, one for each member of the team. These individual reports should contain:

- The design documentation (truth tables, Boolean expressions, karnaugh maps, sub-circuits, etc.) for the individual contribution to the final circuit. This should include a written description of how your component operates and interacts with the rest of the circuit.
- A realistic assessment of your own overall contribution to the assignment. You should give yourself a mark out of 10, where 0 is considered to be very poor (no real contribution) and 10 is outstanding (you did it all). You must use one or two paragraphs to justify your self-assessment.
- An assessment of the other members of your team. You should include a summary of their work, and any comments you may have on their work (good and bad). You should mark your colleagues out of 10, justifying your mark in your review. Note that your colleagues should not see this report.
- A critical review of the assignment itself. What have you learned by attempting this assignment. This should be no more than two or three paragraphs.

You will be required to demonstrate your system as a group in the first seminar after the hand-in date. It is expected that the demonstration should last no more than five minutes per group. Each member of the group is expected to contribute by describing those parts that they designed. Demonstrations should be carried out in an informal, but professional manner.

Marking Scheme

Group Report	**Marks**	**Individual Report**	**Marks**
Design Approach	10	Design Documentation	20
Design of final circuit	20	Self Assessment	10
Group Demonstration	10	Peer Assessment	10
Testing	10	Critical review of assignment	10

Signatures

Lecturer .....

Quality Assessor .....